ECA1803 – OPTOELECTRONICS AND OPTICAL COMMUNICATION

CO3 SCENARIO BASED ASSIGNMENT

Design and Performance Evaluation of an Integrated Photodetection, Imaging, Biomedical Sensing and Display System

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Assumptions

The assignment does not give a few values that are needed for the calculations, so I have assumed the following. They are used throughout the report.

- PPG uses an 850 nm LED with a silicon PIN photodiode, so R = 0.60 A/W is reasonable at this wavelength.
- LiDAR uses a 1550 nm (eye-safe) laser with an InGaAs APD. The given R = 0.80 A/W is the unity-gain (M = 1) responsivity.
- Dark current Id = 1 nA, detector area = 1 mm² (for NEP and D*), temperature T = 300 K, APD excess-noise exponent x = 0.7 (InGaAs).
- Equivalent load of the receiver RL = 1 kΩ and noise bandwidth B = 10 MHz for the SNR model; for the RC-limit calculation C = 0.5 pF and RL = 50 Ω.
- The given signal (−22 dBm) and noise (−37 dBm) are taken as a power ratio.
- PV cell area is 243 cm² (156 mm × 156 mm) under 1 sun (1000 W/m²).

Stage 1 – Interpret the Application Requirements

- **PPG:** the detector has to pick up a very small AC change (microamp level) sitting on a large DC level. It needs good responsivity at 850 nm, low noise and high linearity.
- **LiDAR:** the returned pulse is only a few μW, so the detector needs internal gain, a bandwidth of several GHz and low timing jitter.
- **Weak-signal detection:** low dark current, low NEP and high detectivity are needed so that the signal is not buried in noise.
- **Responsivity, quantum efficiency and detectivity:** responsivity decides how much current we get per watt, quantum efficiency (η) shows how many carriers are generated per photon, and detectivity compares detectors of different areas.
- **NEP:** it is the smallest power that gives SNR = 1 in 1 Hz bandwidth. A smaller NEP means better low-light detection.
- **Linearity:** the photocurrent must be proportional to the optical power over the whole range, otherwise PPG amplitude and image brightness will be distorted.
- **Speed:** the response time is limited by the carrier transit time and the RC time constant, so both have to be kept small for high-speed detection.
- **CMOS/CCD sensors:** high quantum efficiency, low read noise, good dynamic range and fast readout. Small pixels give high resolution but collect less light, and SoC integration reduces size and power.

- **Flexible PV:** thin, lightweight, bendable, with reasonable efficiency and good mechanical reliability for wearables.
- **Displays:** OLED, MicroLED and quantum-dot displays need high brightness, high contrast, fast response, wide colour gamut and long life. AR/VR/MR needs very high brightness and pixel density, and foldable displays need mechanical flexibility.
- **Practical constraints:** power consumption (battery life), heating, ease of integration, reliability and complexity of the circuit.

Application Requirement Statement

The platform must detect microamp-level PPG changes and low-microwatt LiDAR returns reliably (SNR well above 15 dB), respond within tens of picoseconds for ranging, capture face and UV information with low-noise CMOS/semiconductor sensors, show information on a bright and efficient display, and run at low power with support from a small PV source, all within a compact, thermally safe and reliable package.

Stage 2 – Investigate and Select the Photodetector Configuration

2.1 Responsivity

The parameter that converts the received optical power into photocurrent is the responsivity R (A/W):

$$R = Iph / P = \eta q\lambda / (hc) \approx \eta\, \lambda(nm) / 1240$$

2.2 PPG photocurrent

$$\Delta Iph = R \times \Delta P = 0.60\ A/W \times 25 \times 10^{-6}\ W = 15 \times 10^{-6}\ A = 15\ \mu A$$

The blood volume in the tissue changes with every heartbeat, so the light reflected back to the detector also changes. The 15 μA variation follows this change. The AC part of the signal gives the pulse waveform, from which heart rate and heart-rate variability are found (and SpO_2 if two wavelengths are used). The DC part depends on skin, tissue and contact.

2.3 PIN versus APD

Parameter	PIN photodiode	APD
Responsivity	R = ηqλ/hc (0.6 A/W here)	M × R, higher because of avalanche gain
Internal gain	None (M = 1)	10 – 100
Sensitivity	Moderate	High, good for weak signals
Noise	Low (shot + dark current)	Higher, extra avalanche (excess) noise F = M^x
Bandwidth	High, limited by transit time and RC	High, but gain–bandwidth product limits it at large M
Bias voltage	Low (about 5 V or less)	High (tens of volts), needs temperature compensation
Complexity	Simple and cheap	Complex bias and control circuit, costly

2.4 LiDAR photocurrent and APD gain

$$Iph = R \times P = 0.80\ A/W \times 3 \times 10^{-6}\ W = 2.4 \times 10^{-6}\ A = 2.4\ \mu A$$

For a PIN diode this 2.4 µA is small compared with the amplifier noise. The APD multiplies the photocurrent by M before the amplifier adds its noise, so with M = 10 the current becomes 24 µA and the signal rises well above the thermal noise of the receiver. That is why an APD is used for weak optical signals.

2.5 Dark current, quantum efficiency, NEP, detectivity and linearity

- **Dark current:** it is the current with no light. It creates shot noise (2qId·B) and sets the minimum detectable power. For Id = 1 nA, the photocurrent equals the dark current when P = Id/R = 1 nA / 0.8 A/W = 1.25 nW, so below this level the detector is dark-current limited.
- **Quantum efficiency:** $\eta = R \cdot hc/(q\lambda) = 0.60 \times 1240/850 = 0.875$ (87.5 %) for the PPG diode at 850 nm, and $\eta = 0.80 \times 1240/1550 = 0.64$ (64 %) for the InGaAs LiDAR detector at 1550 nm. Higher η means more photocurrent for the same light.
- **NEP:** NEP = in/R. Taking only dark-current shot noise, $in = \sqrt{(2qId)} = \sqrt{(2 \times 1.602\times10^{-19} \times 10^{-9})} = 1.79\times10^{-14}$ A/√Hz, so NEP = 1.79×10^{-14} / 0.60 = 2.98×10^{-14} W/√Hz. For B = 10 MHz the minimum detectable power is about $2.98\times10^{-14} \times \sqrt{(10^7)} \approx$ 94 pW.
- **Detectivity:** D = 1/NEP = 3.35×10^{13} Hz^½/W, and D* = D√A = $3.35\times10^{13} \times \sqrt{0.01}$ cm² = 3.35×10^{12} cm·Hz^½/W (Jones). These values are for the dark-current-limited case; the real receiver is worse because of thermal noise.
- **Linearity:** the photocurrent should follow R·P over several decades of power. Good linearity keeps the PPG waveform undistorted and gives correct pixel values in imaging. PIN diodes are very linear; APDs are linear only up to the point where space-charge and saturation start at high gain.

Design Decision 1: I recommend a **silicon PIN photodetector for PPG and an InGaAs APD (M ≈ 10) for LiDAR and low-light detection** because the PPG signal (15 µA) is already large enough and needs low noise, linearity and a simple low-voltage supply, while the LiDAR signal is only 2.4 µA and the APD raises it to 24 µA, which keeps the signal above the receiver noise and gives a bandwidth of several GHz.

Stage 3 – Noise and SNR Strategy

3.1 SNR from the given powers

The performance metric that compares signal and noise is the signal-to-noise ratio:

$$SNR(dB) = Ps(dBm) - Pn(dBm) = -22 - (-37) = 15\ dB$$

As a ratio, Ps = 10^(−2.2) mW = 6.31 µW and Pn = 10^(−3.7) mW = 0.199 µW, so SNR = 6.31/0.199 = 31.6 (10 log 31.6 = 15 dB).

3.2 Noise mechanisms

- **Shot noise:** random arrival of photons and carriers. $i^2_{shot} = 2q(I_p + I_d)\cdot B$. It increases with the square root of the signal.
- **Dark-current noise:** shot noise of the dark current, $i^2_d = 2q\cdot I_d\cdot B$. It is present even without light and becomes the main limit at very low power.
- **Avalanche noise:** the multiplication process is random, so the shot noise is amplified by M^2F, $i^2_{APD} = 2q(I_p + I_d)\cdot M^2\cdot F\cdot B$, with $F = M^x$.
- **Thermal noise:** $i^2_{th} = 4kT\cdot B/R_L$ from the load resistor, which is independent of the gain.

3.3 Effect of APD gain on noise and SNR

The signal grows as M^2 but the shot noise grows as $M^{(2+x)}$, so SNR first rises (while thermal noise dominates) and then falls (when avalanche noise dominates). Using the model in Stage 7, at P = 3 μW the optimum is $M \approx 4.6$ and at 0.1 μW it is $M \approx 16$. At the design point P = 3 μW, B = 10 MHz and M = 10, the noise currents are: shot 62 nA, thermal 12.9 nA, so total ≈ 63 nA and $SNR = (24\ \mu A)^2/(63\ nA)^2 = 143 = 51.6$ dB.

3.4 Dominant noise in low light

When the received power falls, the signal current $R\cdot P$ falls but the dark current stays constant. Once $R\cdot P$ approaches I_d (about 1.25 nW here) the dark-current shot noise and the thermal noise of the amplifier dominate and the SNR drops quickly. At moderate power (μW level) with gain, the avalanche-multiplied shot noise is the main term. Noise power also grows linearly with bandwidth, so reducing B from 100 MHz to 10 MHz and 1 MHz raises SNR by 10 dB each step (41.6 → 51.6 → 61.6 dB in the simulation).

3.5 Methods to improve SNR

- **Suitable gain:** use M near the optimum (5 – 15), not the maximum.
- **Optical filtering:** a narrow band-pass filter at the source wavelength removes sunlight and ambient background.
- **Reduce dark current:** use a small active area, good passivation and a low bias margin.
- **Cooling:** dark current roughly doubles for every 8 – 10 °C rise, so even a few degrees of cooling helps.
- **Bandwidth optimisation:** use only the bandwidth the signal needs (about 10 Hz for PPG, a few GHz for LiDAR).
- **Better optical coupling:** lenses or concentrators focus more light onto the detector, which raises the signal without extra noise.

Design Decision 2: I recommend an **APD with moderate gain (M ≈ 10) combined with optical filtering, a small-area low-dark-current detector, mild cooling and bandwidth limiting** because the given SNR of 15 dB is only just below the 15.6 dB needed for a bit-error rate of 10^{-9} ($Q = 6$, $SNR = Q^2 = 36$), whereas this strategy gives about 51.6 dB at 3 μW and still about 34 dB at 0.1 μW, so low-light detection is reliable.

Stage 4 – Response Time, Bandwidth and High-Speed Detection

The parameter that relates transit time to bandwidth is the 3 dB bandwidth $f \approx 0.44/tr$.

$$f = 0.44 / tr = 0.44 / (20 \times 10^{-12}\ s) = 22 \times 10^{9}\ Hz = 22\ GHz$$

- **Carrier transit time:** electrons and holes need time to cross the depletion region, so the output current lags the light. A thinner depletion layer gives a shorter transit time but a higher capacitance and lower quantum efficiency.
- **RC time constant:** the junction capacitance C and the load resistance form a low-pass filter with $f = 1/(2\pi RC)$. For C = 0.5 pF and RL = 50 Ω, $f = 1/(2\pi \times 50 \times 0.5\times10^{-12}) =$ 6.4 GHz. Combining both limits: $1/f^2 = 1/22^2 + 1/6.4^2$ gives **$f \approx$ 6.1 GHz**, so the RC limit is the one that controls the speed.
- **Capacitance:** $C = \varepsilon A/d$, so a larger area gives larger capacitance and lower bandwidth. For example, C = 2 pF gives only 1.6 GHz with 50 Ω.
- **Trade-off:** large area gives easier optical coupling and more collected light (higher sensitivity) but more capacitance and less bandwidth. Small area gives high speed but needs good focusing optics. A diameter of about 50 – 100 µm is a good compromise for LiDAR.
- **Receiver circuit:** a TIA adds its own input capacitance and a feedback pole, so the overall bandwidth is lower than the detector bandwidth. A low-capacitance, low-noise TIA placed very close to the detector is needed.

For ranging, the rise time is about $0.35/f = 0.35/6.1$ GHz ≈ 57 ps, giving a range resolution $\Delta r = c \cdot t/2 = 3\times10^{8} \times 57\times10^{-12} / 2 \approx 8.6$ mm, which is enough for automotive LiDAR.

Design Decision 3: I recommend a **small-area (about 50 µm) InGaAs APD with a low-capacitance TIA** as the high-speed detector configuration because the transit-time limit is 22 GHz, the RC limit is 6.1 GHz with C = 0.5 pF, so the small area keeps the capacitance low and the bandwidth in the GHz range, and the lens-based coupling makes up for the loss in collected light.

Stage 5 – PV, Image-Sensor and Emerging Display Technologies

A. Solar Photovoltaic Analysis

$$Pmax = Voc \times Isc \times FF = 0.60\ V \times 5\ A \times 0.75 = 2.25\ W$$

For the assumed cell area of 243 cm², Pin = 1000 W/m² × 0.0243 m² = 24.3 W, so efficiency η = Pmax/Pin = 2.25/24.3 ≈ 9.3 %.

- **Voc and Isc:** Voc is the voltage at zero current (maximum voltage on the I–V curve) and Isc is the current at zero voltage (maximum current). Their product Voc·Isc is the ideal upper limit of power, and the real maximum power point lies below it.
- **Fill factor:** FF = Pmax/(Voc·Isc) shows how square the I–V curve is. It is affected by series and shunt resistance. A higher FF means higher power for the same Voc and Isc.
- **FF and efficiency:** η = Voc·Isc·FF / Pin, so efficiency is directly proportional to FF.
- **Flexible PV for wearables:** suitable because it is thin, light and can bend with clothing or straps.

- **Advantages:** low weight, conformal fitting, extends battery life, can be integrated into fabric or housing.
- **Limitations:** lower efficiency than rigid cells, small usable area in a wearable, output depends on lighting, and encapsulation and bending reduce lifetime. It can therefore supply only auxiliary power.

B. CMOS / CCD Image Sensors

Parameter	CCD	CMOS
Architecture	Photodiodes/MOS capacitors with charge-transfer registers; single output amplifier	Photodiode and amplifier in every pixel; row/column addressing
Pixel operation	Charge is collected and shifted pixel to pixel	Charge is converted to voltage inside the pixel
Readout	Serial, through one or a few output nodes	Parallel column-wise, random access possible
Power consumption	High (many clock drivers, high voltages)	Low (single low supply)
Speed	Limited by serial transfer	High frame rates, windowing
Noise	Very low read noise, good uniformity	Earlier higher fixed-pattern noise, now comparable with correlated double sampling
Pixel scaling	Difficult, needs special process	Easy to scale with standard CMOS (below 1 μm pixels)
SoC integration	Not possible on the same chip	ADC, ISP and logic on the same chip
Face recognition	Good image quality but bulky and power-hungry	Preferred: fast, low power, on-chip processing, NIR-enhanced versions
Automotive / imaging	Rarely used now	Widely used (high dynamic range, global shutter, low power)

When the pixel size is reduced, resolution and sensor density increase and the optics and module become smaller. However, each pixel collects fewer photons, so the signal and full-well capacity decrease, SNR and dynamic range fall, and crosstalk increases. Back-side illumination, micro-lenses and stacked designs are used to compensate. Smaller pixels also allow more circuitry (ADC, memory, processing) to be placed on the same chip, which helps SoC integration but increases heat.

C. Emerging Displays

Parameter	OLED	MicroLED	Quantum-dot (QD)
Light generation	Electroluminescence of organic layers	Electroluminescence of inorganic GaN LEDs	Photoluminescence of QDs (blue backlight) or electroluminescence (QLED)
Brightness	Medium (about 1000 nits)	Very high (over 10^5 nits)	High
Contrast	Very high (true black)	Very high (true black)	High with local dimming

Parameter	OLED	MicroLED	Quantum-dot (QD)
Response speed	Microseconds	Nanoseconds	Fast for QLED, LCD-limited when used with a backlight
Efficiency	Good	Very good	Good, with colour-conversion loss
Colour performance	Wide gamut	Wide, needs colour conversion for RGB	Best, very narrow emission (widest gamut)
Lifetime	Limited (blue ages, burn-in)	Long, stable	Good, QD stability is improving
Power consumption	Low for dark content	Low	Moderate
AR/VR/MR	Usable, brightness limited	Best (brightness, small pixels)	Used as colour-conversion layer
Foldable	Best (thin, flexible)	Possible but transfer is difficult	Possible with flexible QLED

Design Decision 4: I recommend a **flexible thin-film PV, a back-illuminated CMOS image sensor and a MicroLED display (with quantum-dot colour conversion, and OLED for the foldable version)** because the PV is light and bendable for wearable use even though it only gives auxiliary power, CMOS gives low power, high speed and SoC integration, and MicroLED gives the high brightness, fast response, long life and small pixel pitch that AR/VR/MR needs.

Stage 6 – Biomedical and Industrial Applications

1. Wearable Biomedical Sensing (PPG)

In PPG, an LED lights the skin and a photodetector measures the reflected or transmitted light. Each heartbeat changes the blood volume, which changes the absorbed light, so the detected intensity varies with the pulse. The small AC change (15 μA here) is separated from the large DC level by filtering.

- **Source:** green (about 525 nm) works well on the wrist, while red/NIR (660 / 850 nm) penetrates deeper and is used for SpO_2. I chose 850 nm as the assumption for R = 0.6 A/W.
- **Detector and responsivity:** Si PIN, R = 0.6 A/W, with a large area to collect the weak reflection.
- **Noise and conditioning:** the TIA, band-pass filter (about 0.5 – 5 Hz), ambient-light cancellation and ADC remove motion and background light.
- **Response time and power:** only a few hundred hertz of bandwidth is needed, so power is mainly the LED current; pulsed LED drive keeps the average power low.

2. NIR Spectroscopy

NIR light (700 – 2500 nm) penetrates tissue and is absorbed by components such as haemoglobin, water and glucose in a wavelength-dependent way. The detector measures the transmitted or diffusely reflected light and the absorption gives the composition.

- **Wavelength selection:** chosen at the absorption peaks of the target substance (for example 760 and 850 nm for oxygenated and deoxygenated haemoglobin).
- **Detector, sensitivity and noise:** Si PIN up to about 1000 nm and InGaAs PIN beyond that, with low dark current and low noise because the absorption changes are small.
- **Filtering and signal processing:** band-pass filters or a dispersive element, lock-in or synchronous detection, and spectral analysis/calibration in the SoC.

3. Automotive LiDAR

- **Low received power and high sensitivity:** the returned light is only about 3 μW, so an APD with gain ($M \approx 10$) is used.
- **Fast response and bandwidth:** several GHz (6.1 GHz calculated) so that short pulses are not stretched.
- **Accurate ranging:** range = ct/2; the ~57 ps rise time gives about 8.6 mm resolution, and timing jitter is kept small with a low-noise TIA and TDC.
- **Low noise:** optical band-pass filter at 1550 nm, and small area to reduce capacitance and background light.

4. Face Recognition

CMOS sensors are preferred over CCD because of their low power, high speed and on-chip processing. NIR illumination (850 / 940 nm) makes recognition work in the dark and with changing ambient light.

- **Resolution and pixel scaling:** about 2 MP with 2 – 3 μm pixels is enough for the face details; smaller pixels lose sensitivity.
- **Sensitivity and illumination:** back-illuminated or NIR-enhanced pixels and active NIR illumination.
- **Response time and power:** a global shutter at about 30 fps with low power, so that recognition happens in a fraction of a second.

5. UV Sensor

UV detection needs a detector that responds to UV (200 – 400 nm) but not to visible light, with very low dark current. Wide-bandgap semiconductors such as SiC, GaN/AlGaN or diamond give visible-blind (or solar-blind) response, low dark current, good linearity and stability under UV exposure. Ordinary Si photodiodes respond to visible light and degrade under UV, so they need filters.

Design Decision 5: I recommend an **application-specific configuration: Si PIN (850 nm) for PPG, Si/InGaAs PIN with filters for NIR spectroscopy, InGaAs APD for LiDAR, NIR-enhanced CMOS for face recognition, and SiC/AlGaN photodiode for UV** because each application has a different signal level, wavelength and bandwidth, and this choice matches the responsivity, noise, speed and imaging requirement of each one while sharing the same SoC.

Stage 7 – Simulation and Performance Verification

Input parameters

Parameter	Value	Parameter	Value
R (PPG PIN)	0.60 A/W	R (LiDAR, M = 1)	0.80 A/W
APD gain M	10 (swept 1 – 100)	Excess-noise exponent x	0.7
Dark current Id	1 nA	Load RL	1 kΩ
Bandwidth B	10 MHz	Temperature T	300 K
Transit time tr	20 ps	Optical power (LiDAR)	3 µW

MATLAB code

```
% CO3 - detector, noise, SNR and speed model
clear; clc; close all;
q = 1.602e-19; k = 1.381e-23; T = 300;
Rp = 0.60; Ra = 0.80; M = 10; x = 0.7;
Id = 1e-9; RL = 1e3; B = 10e6; tr = 20e-12;

%% A. Responsivity model
P = linspace(0, 50e-6, 200);
figure; plot(P*1e6, Rp*P*1e6, P*1e6, Ra*P*1e6, P*1e6, M*Ra*P*1e6, "--");
xlabel("Optical power (uW)"); ylabel("Photocurrent (uA)"); grid on;
legend("PPG PIN","LiDAR PIN","LiDAR APD M=10");
I_ppg = Rp*25e-6
I_lidar = Ra*3e-6

%% B. Noise vs gain (P = 3 uW)
Mv = logspace(0, 2, 200); Ip = Ra*3e-6;
i_sh = sqrt(2*q*Ip*Mv.^(2+x)*B);
i_d  = sqrt(2*q*Id*Mv.^(2+x)*B);
i_th = sqrt(4*k*T*B/RL)*ones(size(Mv));
i_t  = sqrt(i_sh.^2 + i_d.^2 + i_th.^2);
figure; loglog(Mv, [i_sh; i_d; i_th; i_t]*1e9); grid on;
xlabel("APD gain M"); ylabel("rms noise (nA)");
legend("shot","dark","thermal","total");

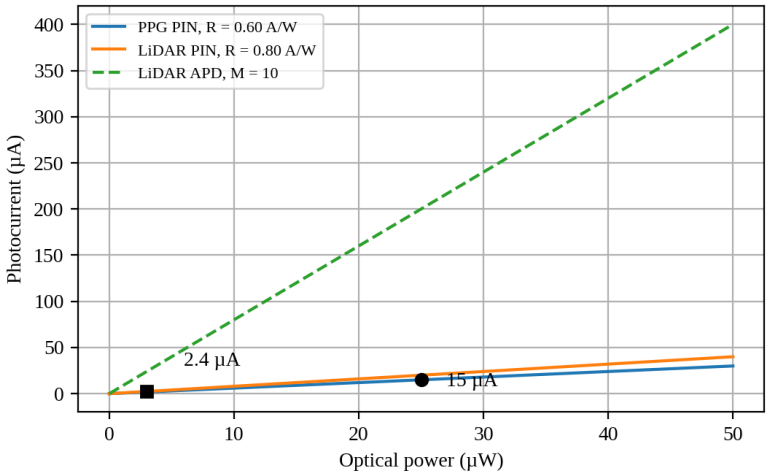
%% C. SNR analysis
snr = @(P,M,B) (M*Ra.*P).^2 ./ (2*q*(Ra*P+Id).*M.^(2+x).*B + 4*k*T.*B/RL);
Pv = logspace(-9, -5, 200); Bv = logspace(5, 9, 200);
figure;
subplot(1,3,1); semilogx(Pv*1e6, 10*log10(snr(Pv,1,B)), Pv*1e6,
10*log10(snr(Pv,10,B)), Pv*1e6, 10*log10(snr(Pv,30,B)));
subplot(1,3,2); semilogx(Mv, 10*log10(snr(3e-6,Mv,B)));
subplot(1,3,3); semilogx(Bv/1e6, 10*log10(snr(3e-6,10,Bv)));
SNR_dB = 10*log10(snr(3e-6,10,B))
SNR_given = -22 - (-37)

%% D. Speed
f_transit = 0.44/tr
f_RC = 1/(2*pi*50*0.5e-12);
f_total = 1/sqrt(1/f_transit^2 + 1/f_RC^2)

%% PV
Pmax = 0.60*5*0.75
```

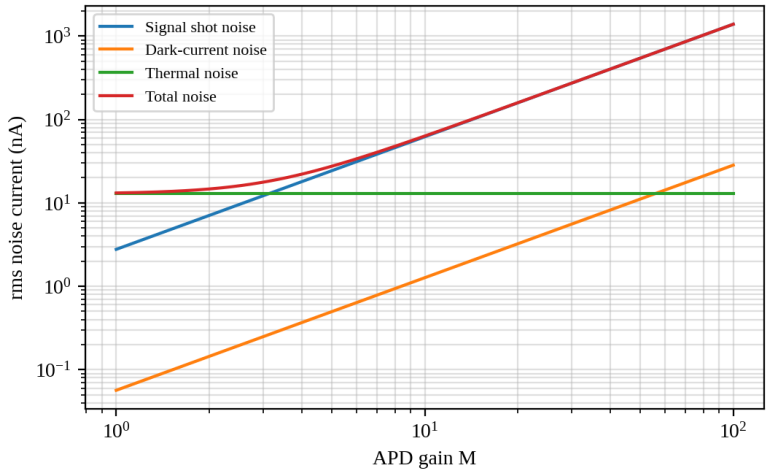
Generated plots

Plot A – Photocurrent versus optical power



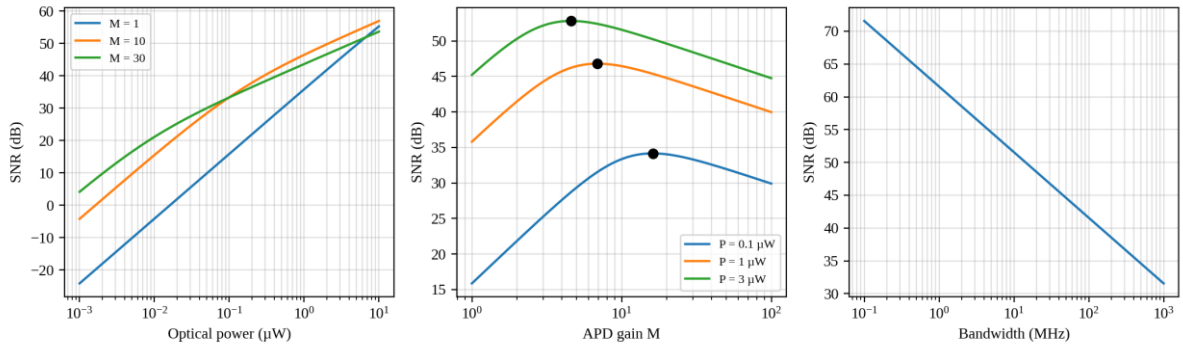
The straight lines show the linear responsivity model; the points mark 15 µA (PPG) and 2.4 µA (LiDAR).

Plot B – Noise components versus APD gain (P = 3 µW, B = 10 MHz)



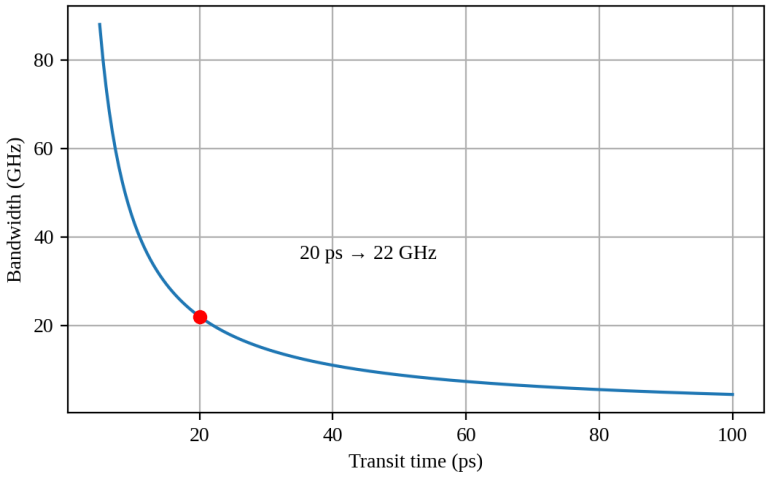
Thermal noise is fixed, shot and dark noise rise with gain, so total noise follows the avalanche terms at higher M.

Plot C – SNR versus optical power, gain and bandwidth



Left: SNR vs power for M = 1, 10, 30. Middle: SNR vs gain with the optimum marked. Right: SNR vs bandwidth (P = 3 µW, M = 10).

Plot D – Transit-time bandwidth



f = 0.44/tr gives 22 GHz at 20 ps.

Theoretical and simulated results

Quantity	Theory (hand calculation)	Simulation	Deviation
PPG photocurrent	15 µA	15.0 µA	0 %
LiDAR photocurrent	2.4 µA	2.4 µA	0 %
Given low-light SNR	15 dB	15.0 dB	0 %
SNR at P = 3 µW, M = 10, B = 10 MHz	51.6 dB	51.56 dB	< 0.1 %
Optimum gain at 3 µW	about 4.6	4.59	< 1 %
Transit-time bandwidth	22 GHz	22.0 GHz	0 %
Combined bandwidth (RC + transit)	6.1 GHz	6.11 GHz	< 0.2 %
PV maximum power	2.25 W	2.25 W	0 %

Interpretation: the simulated values match the hand calculations because the same equations are used, so differences are only rounding. The model shows that gain helps only up to an optimum (M ≈ 5 – 16 depending on power), and that SNR drops 10 dB per decade of bandwidth. In a real receiver the results will deviate because of non-ideal excess noise, amplifier noise, temperature-dependent dark current, and gain–bandwidth limits, so a prototype test is still needed.

Final System Verification

Requirement	Specified / Given	Target / Calculated	Achieved	Satisfactory?
PPG photocurrent response	R = 0.6 A/W, ΔP = 25 µW	15 µA	15 µA	Yes
LiDAR photocurrent	R = 0.8 A/W, P = 3 µW	2.4 µA (24 µA with M = 10)	24 µA	Yes
Detector responsivity	0.60 and 0.80 A/W	As given	0.60 / 0.80 A/W	Yes

Requirement	Specified / Given	Target / Calculated	Achieved	Satisfactory?
Quantum efficiency	$\eta \leq 1$	87.5 % (850 nm), 64 % (1550 nm)	Same	Yes
NEP / Detectivity	Low NEP	2.98×10^{-14} W/√Hz; D* = 3.35×10^{12} Jones	Same (dark-limited)	Yes
Low-light SNR	−22 dBm / −37 dBm	15 dB	15 dB (51.6 dB with APD at 3 µW)	Yes, marginal without gain
Noise performance	Low	Shot-dominated with gain	Optimum M ≈ 5 – 16	Yes
High-speed bandwidth	tr = 20 ps	22 GHz (transit)	6.1 GHz (with RC)	Yes
PV maximum power	Voc 0.6 V, Isc 5 A, FF 0.75	2.25 W	2.25 W (≈ 9.3 %)	Yes
Image-sensor selection	CMOS / CCD	CMOS (BSI)	CMOS	Yes
Display selection	OLED / MicroLED / QD	MicroLED (+ QD, OLED foldable)	MicroLED	Yes
Biomedical sensing	PPG and NIR	Si PIN / InGaAs PIN	15 µA PPG signal	Yes
LiDAR sensing	Fast, weak signal	APD, 6.1 GHz, 8.6 mm	Same	Yes
Automotive sensing	LiDAR and vehicle sensing	LiDAR + CMOS camera	Same	Yes
Face recognition	CMOS/CCD + NIR	CMOS with NIR illumination	Same	Yes
UV sensing	UV detection	SiC / AlGaN photodiode	Same	Yes
MATLAB/SCILAB simulation	Models and plots	Plots A – D	Matches theory	Yes
Overall platform	All constraints	Integrated	Meets requirements (prototype validation pending)	Yes

The calculated and simulated values agree with each other and with the application requirements. The only weak point is the 15 dB low-light SNR without gain, which is marginal and is fixed by using the APD and the noise-control methods described.

Final System Design – Integrated Optoelectronic Platform

Final parameter / design table

Block	Selected configuration	Key values
PPG detector	Si PIN, 850 nm LED	R = 0.60 A/W, η = 87.5 %, ΔI = 15 µA
LiDAR detector	InGaAs APD, 1550 nm, 50 µm	R = 0.80 A/W, η = 64 %, I = 2.4 µA (24 µA at M = 10)

Block	Selected configuration	Key values
NEP / detectivity	PIN, Id = 1 nA, 1 mm²	NEP = 2.98×10⁻¹⁴ W/√Hz, D* = 3.35×10¹² Jones
Low-light SNR	APD + filter + mild cooling	15 dB (given); 51.6 dB at 3 µW, M = 10, 10 MHz
Bandwidth	Low-capacitance TIA	22 GHz transit, 6.1 GHz with RC (C = 0.5 pF)
PV	Flexible thin-film	Pmax = 2.25 W, η ≈ 9.3 %
Image sensor	Back-illuminated CMOS, global shutter	NIR-enhanced, SoC integrated
Display	MicroLED + QD colour conversion (OLED foldable)	High brightness, ns response
NIR spectroscopy	Si/InGaAs PIN with band-pass filters	760 / 850 nm and longer bands
Face recognition	CMOS + 850/940 nm NIR illuminator	About 2 MP, 2 – 3 µm pixels
UV sensor	SiC / AlGaN photodiode	Visible-blind, low dark current
Simulation	MATLAB model	Matches theory

System block diagram

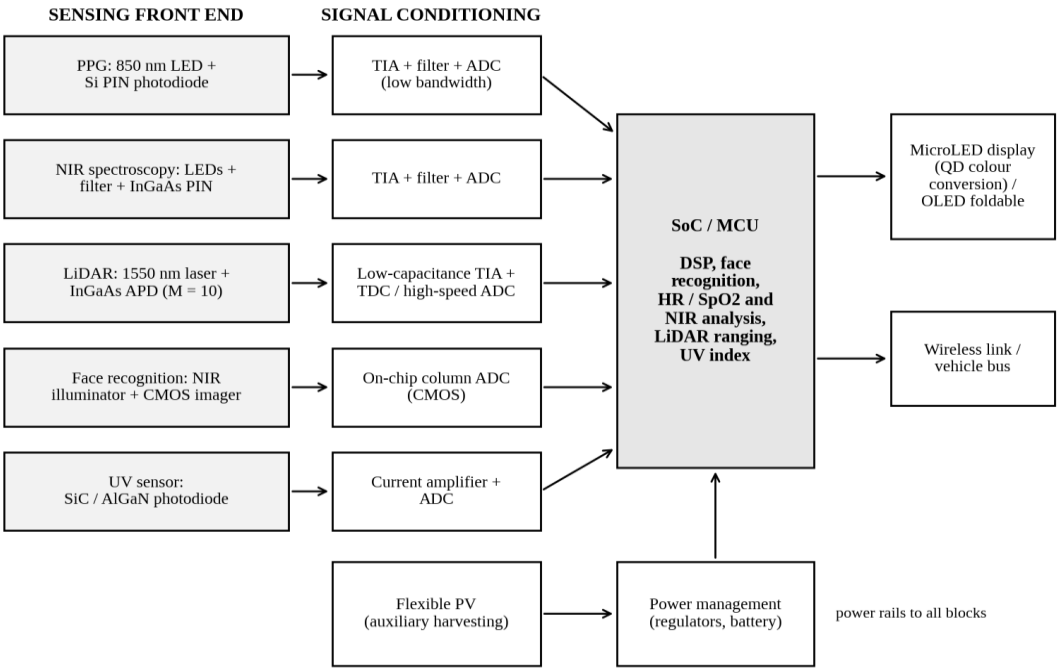


Figure: Block diagram of the integrated optoelectronic platform

Summary of the final design

The platform uses a Si PIN diode for PPG (15 µA signal), an InGaAs APD with M = 10 for LiDAR (2.4 µA raised to 24 µA, bandwidth about 6.1 GHz) and filtered PIN detectors for NIR spectroscopy. A back-illuminated CMOS sensor with NIR illumination handles face recognition and a SiC/AlGaN diode handles UV. Noise is controlled by moderate gain, filtering, low dark current, limited bandwidth and good coupling, giving 51.6 dB SNR at 3 µW.

A 2.25 W flexible PV source gives auxiliary power and a MicroLED display is used for the output. MATLAB simulation agrees with the theoretical values.

Final Engineering Decision

Would I deploy the integrated optoelectronic platform? YES, with the condition that a prototype test is done first.

- **Detector, responsivity and η:** the PIN (0.6 A/W, 87.5 %) and APD (0.8 A/W, 64 %) choices give the required photocurrents of 15 μA and 2.4 μA (24 μA with gain).
- **NEP and detectivity:** NEP of about 3×10^{-14} W/√Hz and D* of about 3.4×10^{12} Jones are good enough for μW-level signals.
- **Noise and SNR:** the given 15 dB is marginal, but the APD with $M \approx 10$ and the noise-control methods give 51.6 dB at 3 μW and about 34 dB at 0.1 μW.
- **Speed:** 22 GHz transit limit and 6.1 GHz with RC give about 8.6 mm range resolution.
- **PV, image sensor and display:** 2.25 W PV is useful as auxiliary power only, CMOS gives low power and SoC integration, and MicroLED meets the AR/VR/MR needs.
- **Applications and simulation:** PPG, NIR, LiDAR, face recognition and UV needs are all met, and the MATLAB results agree with the hand calculations.
- **Integration, power and reliability:** the APD bias supply, temperature control of the APD, MicroLED manufacturing cost and flexible-PV lifetime are the main risks, so field tests across temperature and ambient light are needed before final deployment.

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